

Further experiments

Synthesis of previous experiments up to current results

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Topics

Key points to talk about

Recap

MIDOG++

Results

- Exact-B
- SCEPTRE-HARP

Conclusion

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Recap

Reduced routing risk through composition

What was established on the Camelyon17 dataset:

- Support-NELBO did not translate reliably into Top-1 downstream selection.
- Dense geometric composition exploited complementary CVAE experts.
- Best dense result: **0.814** BACC in the previous experimental setting.

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Last time, I found that compatibility estimation alone was not sufficient. Selecting one expert was brittle, while combining several frozen experts recovered complementary downstream signal. But composition only avoids a potentially wrong choice—it does not solve the routing problem.

Recap

But it did not solve routing

What remained unsolved:

- Composition avoids committing to one potentially wrong expert.
- It does not identify which source expert is compatible with an unseen domain.
- The previous setting offered limited independent domains for rigorous LODO evaluation.
- Move to MIDOG++, with scanner centers as explicit domains.

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I therefore moved to MIDOG++, where scanner centers provide explicit domains for leave-one-domain-out evaluation.

MIDOG++ dataset

Provides more explicit domains for LODO evaluation

Key dataset numbers:

- 492 cases
- 10 observed domains
- 9 eligible LODO domains
- 22.569 image patches

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MIDOG++ dataset

What the dataset contains

- Histopathology data from human and canine tumors
- 7 tumor entities, collected at 4 institutions/origins
- Images acquired using 4 scanner models
- Binary task: mitotic figure vs hard-negative/non-mitotic
- Thesis contract: 11.937 mitotic and 10.632 negative patches

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MIDOG++ dataset

How it is used in this thesis

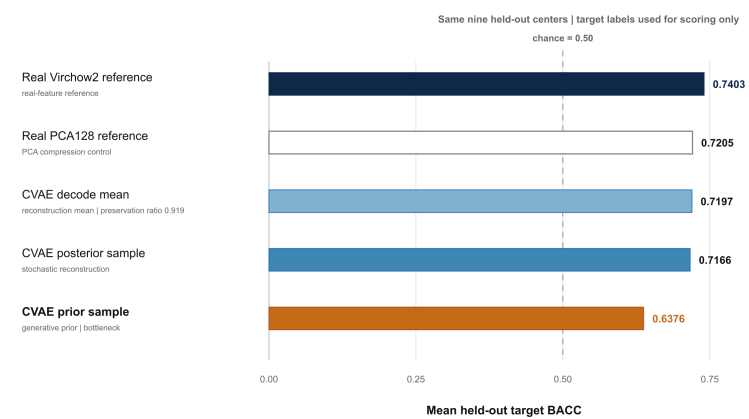
- Each sample is an annotation-centered 224 x 224 patch
- Patches are represented by frozen 3.840-dimensional Virchow2 embeddings
- Domain definition: tumor type x laboratory/origin x scanner model
- Case-disjoint 45% / 10% / 45% train-validation-test split
- In every LODO fold: 8 source domains → 1 unseen target domain
- One small domain is quarantined, leaving 9 eligible evaluation domains

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MIDOG++ turns routing into a repeated unseen-domain problem: can frozen source experts be selected or composed for a target domain that was unavailable during expert training?

Preservation bottleneck

Reconstruction preserved task signal, but prior generation did not

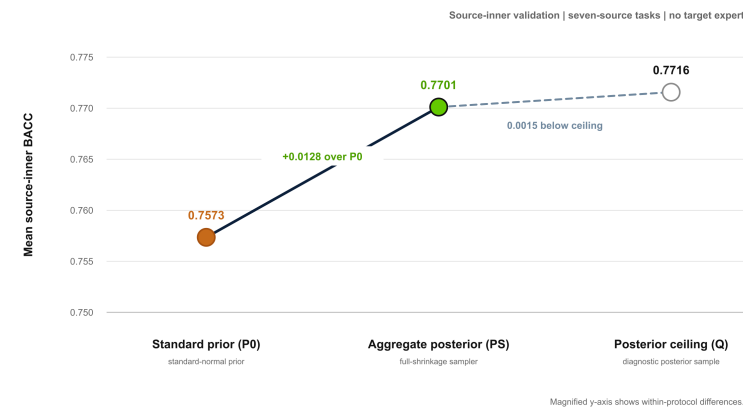


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The CVAE could reconstruct useful embeddings, but its standard prior did not generate equally useful samples. Prior recovery therefore had to be solved before routing.

Bundled redesign

System redesign restored generative utility



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The original result was not evidence that the CVAE fundamentally failed.

A bundled redesign of representation capacity, architecture, training duration, latent capacity, and classifier regularisation raised the standard-prior baseline from 0.602 to 0.757 BACC.

With that system fixed, aggregate-posterior sampling increased performance to 0.770 and closed approximately 90% of the remaining posterior-ceiling gap.

GenerationLock

Ensures that only the routing decision changes

Frozen component	Setting
Candidate pool	8 non-target source domains
Generation budget	128 samples/source/class
Total budget	1.024 samples/class
Replication	3 training × 3 generation seeds
Source models and priors	Frozen
Target data	Not used
Target labels	Final scoring only

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With experts, priors, seeds and budgets frozen, Exact-B becomes the controlled reference against which routing must improve.

Results

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Appendix

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A1: Uniform-B redesign

The performance increase is a system-level result

Component	Earlier Configuration	Uniform-B v2
Representation	PCA128 (96+32)	PCA256 (192+64)
Hidden architecture	2 × 512	3 × 1024
Latent dimension	32	64
Training duration	1.000 steps	4.000 steps
Batch size	128	256
Learning rate	0.001	0.0005
Synthetic composition	Single course, 256/class	Seven-source equal union, 1.024/class
Classifier régularisation	C=1.0	C=0.01
Prior evaluation	P0 and posterior-ratio Pq	P0, aggregate PS, posterior Q/QM

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These changes were introduced jointly. The performance increase is therefore a system-level result, not evidence for the causal contribution of any individual change.